

S-Plane detection — audit, corrections, and final validated results

Date: 2026-09-20 · **Revised:** 2026-09-29 (two claims corrected — see footer) · **Status:** corrected and re-validated · **Supersedes** all earlier 2026-09-20 result sheets

Why this document exists

A cross-check audit of the same day's work found **four defects, two of them fabricating data**.

Everything below is the corrected position. Earlier numbers issued on 2026-09-20 (v2 `sensitivity 1.000 [0.962, 1.000]`, `specificity 0.800`, `C1/C2/C3 12/12`) are **withdrawn**. Nothing is hidden; each defect is stated with the evidence that proved it.

The four defects

D1 — 36 "replicates" were one capture copied 12 times (*fabricated data*)

`run_gap.sh` was created mode **644** but invoked as `./run/run_gap.sh`, returning **exit 126 (Permission denied)**, silenced by `>/dev/null 2>&1`. The packet-count guard then read a **stale** capture left in the working directory, passed it, and copied it into all 12 replicate folders.

Proof: all 12 captures per C-scenario had **one distinct SHA-256**; the three scenarios completed in **≤1 second** per rep in the campaign log, where a genuine run takes ~48 s. *Fix:* permissions corrected; runner now invoked via `bash`, **exit code checked**, and the working directory **purged before every run** so a stale capture can never satisfy the guard. All 36 runs re-executed.

D2 — `B3_pdv_congestion` never injected anything (*fabricated scenario, also affects the earlier campaign*)

This kernel has no `sch_netem` (`tc ... netem` → "Specified qdisc kind is unknown"), and `scenarios.sh` swallowed the failure with `2>/dev/null || true`. B3 was a **baseline run wearing a congestion label**.

Proof: B3 PDV was statistically identical to baseline (ru1 median 62.4 ms / sd 36.0 ms vs 61.7 / 35.5), and identical to the untouched ru3 control. *Fix:* replaced with a **physical congestion model** — a `tbf` bottleneck on the BC → RU timing path plus competing **non-PTP** background traffic through it, so PTP queues behind data-plane traffic. Deep queue ⇒ packets are *delayed, not dropped*. Sync PDV now rises from **0.07 ms to 5.6–857 ms**. All 12 reps re-run. **This defect also invalidates B3 in the earlier 132-run campaign cited in the submitted documents.**

D3 — the v2 detector produced false positives and destroyed an honest abstention

v2's rate-starvation detector (D1) had **no minimum-sample and no transient guard**. A boundary clock that merely appears briefly during BMCA start-up (3–5 frames) was read as "rate starved."

Proof (upstream captures): `B2_gm_failover` **BENIGN** → **ATTACK**; `B_unplanned_failover` **UNKNOWN** → **ATTACK**, i.e. it overwrote the deliberate abstention. Trigger: 3 frames → 0.06 Hz vs 16 Hz declared. This is the exact failure the frozen base rule had already solved with `TRANSIENT_FRACTION`; v2 reintroduced it. *Fix:* **decision_rule_v3.py** (frozen aa9417b7) adds a minimum-frame floor and a

persistence requirement (a source must be active across $\geq 50\%$ of the window before its rate is judged), includes the provisioned *replacement* BC identity, and requires a UTC-offset change to be sustained. v3 fixes both false positives and is **verdict-identical to v2 on every downstream capture** — it removes false positives without altering a single true positive.

D4 — the evaluator overstated and under-checked

`evaluate_v3.py` pooled metrics in a way that moved with run count, "proved" additive-only by comparing two scalars, credited abstention even on error paths, had no replicate whitelist, and silently dropped attribution scoring. Replaced by `evaluate_v4.py`: per-scenario primary metrics, macro-averages, per-run additive-only verification, substantive-path-only abstention, explicit rep whitelist, restored attribution, and a fully enumerated run list written into the artifact.

Disclosure — frozen-manifest changes

`run/randparams.py` and `run/scenarios.sh` no longer match the 2026-09-17 manifest (`verify_frozen.sh` will flag them). Reasons are recorded in `FROZEN_HARNESS_EPOCH2.json`. **The decision rules themselves are unchanged and still match their original hashes** — only the experiment apparatus changed.

Corrected results — 168 live runs (12 reps $\times$ 14 scenarios), all three rules on the same runs

Per-scenario is the **primary** metric. Pooled figures are descriptive only: because both rules are bimodal, a pooled number moves with how many runs of each scenario exist, not with rule quality.

Scenario	Expect	Frozen rule	v3 (primary)	95% CI (v3)
A1 rogue master	ATTACK	12/12	12/12	[0.757, 1.000]
A2 sync spoof	ATTACK	12/12	12/12	[0.757, 1.000]
A3 replay	ATTACK	12/12	12/12	[0.757, 1.000]
A5 DoS flood	ATTACK	12/12	12/12	[0.757, 1.000]
A8 rogue BC	ATTACK	12/12	12/12	[0.757, 1.000]
C1 packet removal	ATTACK	0/12	11/12	[0.646, 0.985]
C2 malformed	ATTACK	0/12	12/12	[0.757,

Scenario	Expect	Frozen rule	v3 (primary)	95% CI (v3)
	K			1.000]
C3 whole-second	ATTACK	0/12	12/12	[0.757, 1.000]
baseline	BENIGN	12/12	12/12	[0.757, 1.000]
B2 GM failover	BENIGN	12/12	12/12	[0.757, 1.000]
B3 congestion (now real)	BENIGN	11/12	11/12	[0.646, 0.985]
B7 topology change	BENIGN	12/12	12/12	[0.757, 1.000]
B_bc_replacement	BENIGN	0/12	0/12	[0.000, 0.242]
B_unplanned_failover	UNKNOWN	12/12	12/12	[0.757, 1.000]

Summary	Frozen rule	v3 (primary)
Macro sensitivity (8 attack scenarios)	0.625	0.990
Macro specificity (5 benign scenarios)	0.783	0.783 (identical)
Abstention correctness (substantive path)	1.000	1.000
Attribution accuracy	0.552	0.875

Additive-only, verified per run (not by scalar): 0 runs where the frozen rule said ATTACK and v3 did not. 35 escalations, all on C1/C2/C3. v2 and v3 agree on every downstream run.

Measured sensitivity limit (this is a real result, not a caveat)

C1 is **11/12**, and the miss is precisely located. D1 fires when delivered rate drops below 0.5× declared:

Extension to the per-scenario results table (pp. 2-3): source, destination, attack devices, consequence

Added 5 Oct 2026. This page extends the table on the preceding page with four columns requested after the PRISM review: **Source** (who or what originates the attack or fault), **Destination** (where it hits in the O-RAN architecture), **Attack devices** (devices that could originate it; for benign faults, the originating devices) and **Consequence**. All pages before and after this one are unchanged from the 29 Sep 2026 revision.

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
A1	A1 rogue master	Internal attacker on the fronthaul timing network posing as a grandmaster: forged/superior Announce data set wins the BMCA [R7384 s3.2.4 rogue master; TR106 T-SPLANE-02/03; TS105 11.1.5.2.2].	O-RU PTP slave port - its BMCA / parent selection on the Open Fronthaul S-plane; every O-RU on the attacker's segment re-parents to the rogue [TS982; MEAS: 36/36 RU logs selected the rogue as best master].	Any device that can send PTP on the fronthaul segment: compromised O-DU (LLS-C1/C2), compromised fronthaul switch or boundary clock, compromised O-RU, or an unauthorised host plugged into the fronthaul Ethernet at the cell site; supply-chain backdoored DU/RU [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: second ptp4l in namespace 'rogue' on brDN.	False time distributed to all downstream O-RUs [R7384 Table 1]. Published impact in one experiment: O-RU crash ~2 s after attack start with switch PTP ports in dynamic role, 5G cell drop, manual reboot [TSAFE]. At stake: cell phase sync < 3 us, TAE 65 ns MIMO [3GPP].
A2	A2 sync spoof	Attacker injecting forged Sync/Follow_Up (false originTimestamp / correctionField) under an off-allow-list identity [R7384 s3.2.2 spoofing].	O-RU PTP slave port - the timestamp content its servo consumes [TS982].	Any host able to transmit on the fronthaul segment: compromised O-RU, O-DU, switch, or an attached rogue host [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: scapy inject.py from the ru3 namespace.	False time / accuracy degradation if accepted [R7384 Table 1]. Measured: no master change in any RU log; visible only in packet evidence [MEAS].
A3	A3 replay	Man-in-the-middle or injector recording and re-sending genuine PTP frames later (stale timestamps, reused sequenceId) [R7384 s3.2.3 replay].	O-RU PTP slave port - sequence continuity and timestamp freshness of the master's messages [TS982].	Device with capture + transmit access on the segment: compromised switch/port mirror, compromised O-RU/O-DU, attached host [TSAFE s4.1 replay]. Testbed: tcpdump + tcpreplay from ru3.	False time / accuracy degradation if stale messages are accepted [R7384 Table 1; TSAFE replay]. Measured: no master change in any RU log; visible as sequenceId regressions [MEAS].
A5	A5 DoS flood	Flood of PTP traffic at the master/slave or segment [R7384 s3.2.9 DoS against the time protocol; TR106 T-SPLANE-01; TS105 11.1.5.1.1, 24.2.1.1].	PTP processing of the master clock and of the O-RU ports on the segment [TR106 T-SPLANE-01; TS982].	Any device able to transmit on the segment, including a compromised O-RU or an attached host [TSAFE s4]. Testbed: flood.py from ru3 (~38,000 records/run vs ~6,000 baseline).	Clock service interrupted / degraded [TR106 T-SPLANE-01; R7384 Table 1 DoS]. Measured: no master change in any RU log at the rates used [MEAS].
A8	A8 rogue BC	Rogue boundary clock inserted in the path, or an authorised BC/TC turned malicious: relays the real GM downstream under its control (campaign: stepsRemoved +1) [R7384 s3.2.4 in MITM position; TR106 T-SPLANE-03].	O-RU parent selection - every O-RU behind the inserted clock [TR106 T-SPLANE-03; MEAS: 36/36 RU logs took the rogue BC as new master].	Compromised boundary clock or fronthaul switch with PTP, or a two-port host bridging both segments [TSAFE s4; TR106 T-FRHAUL-02]. Testbed: two-port ptp4l in namespace 'rbc'.	Attacker controls the time relayed to all downstream O-RUs; slaves given inaccurate time [TR106 T-SPLANE-03; R7384 Table 1 false time].
C1	C1 packet removal	Man-in-the-middle selectively intercepting and dropping PTP packets [R7384 s3.2.5; TR106 T-SPLANE-04; TS105 11.1.5.3.1].	Downstream port of the boundary clock -> the O-RUs it serves (delivery of Sync/Announce) [TS982].	In-path element: compromised fronthaul switch or boundary clock; external MITM on the link also possible [R7384 s3.1, Table 1]. Testbed: BC downstream port blackholed.	Accuracy loss or free-running / DoS [TR106 T-SPLANE-04]. Measured: 36/36 RU ports lost master on Announce timeout; servo reports 7.8 vs 22 per log [MEAS].
C2	C2 malformed	Injection of structurally invalid PTP frames (fuzzing) to exploit parser weaknesses [R7384 s3.2.11; TS105 24.2.1.2].	PTP parser / stack of the O-RU and other clocks on the segment [TS105 24.2.1.2].	Any host able to transmit on the segment, including a compromised O-RU [TSAFE s4]. Testbed: inject_malformed.py from ru3.	Implementation-dependent: from silent discard to stack crash / DoS [R7384 s3.2.11]. Measured: no master change in any RU log [MEAS].
C3	C3 whole-second	Forged Announce abusing time-properties fields: leap61 set, currentUtcOffset 0 instead of 37, traceability stripped [R7384 s3.2.2 spoofing].	O-RU time-properties data set (UTC offset / leap handling) [TS982].	Any host able to transmit on the segment under the GM's identity [TSAFE s4]. Testbed: inject_wholesecond.py from ru3.	False time by whole seconds in UTC-derived timing [R7384 Table 1 false time]. Measured: no master change; servo reports 3.0 vs 22 per log [MEAS].
baseline	baseline	None (healthy control).	None.	None.	None - reference behaviour.
B2	B2 GM failover	Operator maintenance: primary grandmaster withdrawn during an open maintenance window - no attacker.	BMCA at every clock; O-RUs re-parent to the allow-listed backup GM [G8275].	Originating devices (benign): GM-A (withdrawn), GM-B (provisioned backup).	Brief re-parenting; time stays traceable to an allow-listed GM. Packet-identical to A1 except for allow-list and maintenance context [MEAS: 12/12 BENIGN].
B3	B3 congestion	Competing data-plane traffic queuing PTP behind it in the fronthaul (congestion) - no attacker [G8261].	PTP packets on the BC -> O-RU path (queueing delay at the bottleneck).	Originating device (benign): congested fronthaul switch/link queue. Testbed: tbf 1 Mbit bottleneck on v-bc-dn plus background traffic.	Packet delay variation rises (measured Sync PDV 0.07 ms -> 5.6-857 ms), offset noise grows; 12/36 RU logs hit the Announce timeout - so a timeout alone does not prove an attack [MEAS; G8261].
B7	B7 topology	Planned topology change: a link bounced / O-RU re-homed during maintenance - no attacker.	The affected O-RU's port state and path (stepsRemoved, path delay) [G8275].	Originating devices (benign): fronthaul link / switch port being reconfigured.	The affected O-RU re-parents after the link returns; stepsRemoved and path delay change consistently with the new topology and no field becomes illegal [MEAS: 12/12 BENIGN].

ID	Fault / scenario	Source (of attack or fault)	Destination (where it hits in the architecture)	Attack devices (possible originating devices)	Consequence (effect)
B_bc_replacement	B_bc_replacement	Planned replacement of the boundary clock during maintenance; replacement identity provisioned - no attacker.	O-RU parent selection (new BC identity in the path).	Originating devices (benign): old BC (removed), provisioned replacement BC.	O-RUs lose the old BC (36/36 RU logs hit the Announce timeout during the swap) and re-parent to the replacement [MEAS]. Structurally identical to A8, which is why the base rule mis-reads it (0/12).
B_unplanned_failover	B_unplanned_failover	Grandmaster crash with no maintenance window open - cause not knowable from packets (failure or attack).	BMCA at every clock; O-RUs re-parent to the provisioned backup GM.	Originating devices: GM-A (failed), GM-B (provisioned backup). Intent unknown.	Re-parenting to an allow-listed GM; time stays traceable. Correct handling is escalation (UNKNOWN), not a guess [MEAS: 12/12 UNKNOWN].

Key to bracketed tags: **R7384** IETF RFC 7384 (2014): s3.1 attacker classes, s3.2 threats, Table 1 impact (false time / accuracy degradation / DoS) · **TR106** ETSI TR 104 106 V3.0.0 (2025-06) O-RAN threat model: s7.2 threat agents; T-SPLANE-01..04; T-FRHAUL-01/02 · **TS105** ETSI TS 104 105 V7.0.0 (2025-06) O-RAN security test specification (test-case IDs) · **TS982** ETSI TS 103 982 V8.0.0 O-RAN architecture: O-RU hosts Low-PHY + RF and terminates Open Fronthaul (cl. 3.1, 6.3.6) · **TSafe** Groen et al., TIMESAFE, ACM TOPS 28(5) 2025 / arXiv 2412.13049v3: s4 access paths and attacks; O-RU crash ~2 s after attack start with switch PTP ports in dynamic role (one observed experiment, not a standardised limit) · **G8275** ITU-T G.8275.1 telecom profile clockClass semantics via Calnex one-page summary: 6 T-GM locked, 7 T-GM holdover in spec, 135 T-BC holdover in spec, 140/150/160 T-GM out of holdover spec, 165 T-BC out of holdover spec, 248 no reference since start-up · **G8271** ITU-T G.8271.1 network time-error budget: 1.5 us at reference point E, 1100 ns network limit at point C (v8 deck slide 7; Calnex budget summary) · **G8261** ITU-T G.8261 packet delay variation (PDV) context · **3GPP** 3GPP TS 38.133 cell phase sync < 3 us (via Ruffini, ATIS 2018); TS 38.104 cl. 9.6.3.2 TAE 65 ns MIMO · **BSI** BSI 5G Risk Analysis: Radio Access Network, v1.0 (2025) - GNSS jamming/spoofing · **1588** IEEE 1588-2019 offset/delay computation (asymmetry error = half the path asymmetry) · **MEAS** Measured in this project (168-run campaign, RU1-RU3 ptp4l logs, 36 logs per scenario) · **B6M** Measured in this project: two-laptop crystal comparison, 2 Oct 2026, outputs/B6_two_machine_2026-10-02/run2/DRIFT_REPORT_run2.md (run 2 ESTABLISHED, +21.013 ppm [20.623, 21.403]; run 1 NOT ESTABLISHED, +20.161 ppm; the 0.852 ppm run-to-run difference exceeds either CI). Consumer laptop crystals against an NTP reference - not telecom-class oscillators · **MTC** this catalogue's own entry for the family.

black hole	observed/ declared	verdict
75 %	0.366	caught
70 %	0.414	caught
65 %	0.458–0.462	caught
60 %	0.505	missed (threshold 0.500)

D1 detects sustained interception that blackholes ≥ 65 % of the post-warm-up injection window and evades at 60 %. The blackhole fraction is of the window *after* the 8 s warm-up; the measured window also contains the warm-up and the restore tail, which is why a 75 % blackhole yields a rate ratio of 0.366 rather than 0.25. The untested bracket is 60–65 %; linear interpolation of the measured ratios puts the threshold crossing at ≈ 60.5 % (62.5 % is the midpoint of that bracket, not an interpolation). Replicate counts per level were 6 / 3 / 2 / 1, so the decisive missed level rests on a single run; the 55 % level offered by `randparams.py` was never drawn and, by the same window arithmetic, would also have evaded. Short or bursty interception averages out and is not detected.

Known limitations, not fixed

1. **B_bc_replacement 0/12 for both rules** — a legitimate, planned boundary-clock swap is called an attack. Context carries a single `expected_bc_identity`. It is a deterministic known gap, not a sampling rate; it is the DOMINANT cause of specificity 0.783 and the only scenario producing false ATTACK calls (12 of them). It is not the only benign miss: B3 congestion abstains on 1 of 12 runs, so macro specificity excluding `B_bc_replacement` is **0.979, not 1.000**. Fixing it needs a BC allow-list + re-freeze.
2. **D3 clause (c) will false-positive on real PRTC hardware** — it flags `clockClass  $\leq$  6` with traceability deasserted, which is normal for many production grandmasters. Not exercised here (testbed GMs advertise 248).
3. **TRUE_TAI_UTC_OFFSET = 37** is correct only until the next leap second; override via `ctx['true_utc_offset']`.
4. **C-scenario design space is small** — randomisation draws from only 4–5 distinct values, so 12 replicates are 12 genuine captures but not 12 independent attack *designs*.
5. **Hardware-only, out of software scope:** GNSS spoof (A6), GNSS jam (A7), delay-attack measurement (A4), holdover (B1), SyncE/EEC (B4), oscillator drift (B6).

Evidence

`splane_campaign_CORRECTED_2026-09-20.tgz` (1096 files: all 168 runs' deep CSVs, three rules' verdicts per run, injection/impairment logs, every script, all freeze records) · `EVALUATION_v4.json` (fully enumerated run list) · `FROZEN_HARNESS_EPOCH2.json` (disclosed manifest changes) · `FROZEN_v3.json` (why v3 exists, stated openly). Reproduce: `python3 run/evaluate_v4.py`.

Two claims in the 2026-09-20 issue were corrected after an independent recount of the raw per-run files:

1. **B_bc_replacement** was described as "the entire reason specificity is 0.783 rather than 1.000". It is not. Recounting the five benign scenarios gives macro specificity 0.7833; excluding B_bc_replacement gives **0.9792**, because B3_pdv_congestion misses one run of twelve. The two failures are also of different kinds: B_bc_replacement produces 12 **false ATTACK** calls, while the B3 miss is an **UNKNOWN** on a benign scenario — an over-cautious abstention, not a false alarm.
2. **The interception evasion boundary was given as "~62 %" and described as an interpolation.** 62.5 % is the midpoint of the untested 60–65 % bracket. Linear interpolation of the measured observed/declared ratios across that bracket puts the threshold crossing at **≈60.5 %**. The blackhole fraction is also of the post-warm-up injection window, not of the whole observation window.

No measured value changed. Every per-scenario count, macro metric, confidence interval and rate ratio in this document was recounted from `cap/*__r*` in `splane_campaign_CORRECTED_2026-09-20.tgz` and reproduced exactly.